Expert Systems With Applications 326 (2026) 132723

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Integrating grey Fourier modeling with Transformer for enhanced short-term metro passenger flow prediction

Di Zhao [a,b], Li Xue [c], Yumin Liu [c], Kailing Li [d,*]

[a] College of Business and Accounting, Henan Open University, Zhengzhou 450061, Henan, China
[b] College of Management, Zhengzhou University, Zhengzhou 450001, Henan, China
[c] School of Management Engineering, Zhengzhou University of Aeronautics, Zhengzhou 450046, Henan, China
[d] College of Economics and Management, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, Jiangsu, China

ARTICLE INFO

Keywords:
Metro passenger flow
Grey fourier model
Transformer
Markov chain
Hybrid prediction

ABSTRACT

Accurate prediction of metro passenger flow is essential for refined management and intelligent scheduling of urban rail transit systems. To address the complex characteristics of short-term passenger flow data, such as strong periodicity, nonlinear fluctuations, and stochastic disturbances, this paper proposes a novel deterministic–stochastic hybrid modeling framework. The core idea is to decompose passenger flow sequences into deterministic long-term evolutionary patterns and stochastic fluctuation components. First, a grey fourier model (GFM) is developed to capture deterministic trends, effectively extracting evolutionary patterns and periodic features. Second, to model stochastic fluctuation that cannot be characterized by the GFM, an attention-based Transformer is employed to identify nonlinear relationships and long-term temporal dependencies. Finally, to bridge deterministic and stochastic modeling, a Markov chain is introduced to represent the state-transition characteristics of residuals, thereby correcting hybrid predictions through discrete state partitioning and transition probability matrix estimation. A case study based on real passenger flow data demonstrates that the proposed framework achieves high prediction accuracy. Ablation experiments verify the effectiveness of each component, while comparative analyses confirm the superior performance of the integrated modeling paradigm. This study offers a novel theoretical framework for metro passenger flow prediction and provides significant practical value for improving operational efficiency.

1. Introduction

As global urbanization accelerates, the metro, characterized by large capacity, high efficiency, and low carbon emissions, has become a critical solution to urban transportation challenges (Lin et al., Jul. (2022)). According to the International Association of Public Transport (UITP), as of 2023, more than 200 cities worldwide have constructed and operate rail transit systems, with a total operational mileage exceeding 20,000 kilometers. Urban metro systems offer high punctuality and strong transport capacity, with peak unidirectional cross-sectional passenger flows reaching 40,000–60,000 passengers per hour (Lin et al., May (2021)), equivalent to the capacity of 20–30 dedicated bus lanes. Nevertheless, with the continuous expansion of metro networks and sustained passenger growth, operational management faces increasingly severe challenges. In this context, accurate passenger flow prediction is a critical prerequisite for refined metro management (Liu et al., Jan. (2020)), operational scheduling (Geng et al., Mar. (2024)), and service quality improvement (Li et al., Jan. (2025)). Therefore, research on short-term metro passenger flow prediction holds both theoretical and practical significance.

Short-term metro passenger flow prediction remains a highly challenging research topic. Passenger flow constitutes a complex time series characterized by nonlinearity, non-stationarity, and stochastic fluctuations. Forecasting models have evolved from traditional statistical methods to advanced deep learning frameworks (Xue et al., Nov. (2023)). Early studies primarily relied on parametric time series models, such as autoregressive integrated moving average (ARIMA), seasonal ARIMA (SARIMA) (Hopfe et al., 2024), and exponential smoothing (Holt-Winters) (Trull et al., Feb. (2020)). While these models are structurally simple and theoretically well grounded, they are mainly suited to linear and stationary data. To address this limitation, classical machine learning approaches, including support vector regression (SVR)

* Corresponding author.
E-mail addresses: zhaodi@haou.edu.cn (D. Zhao), xuelitt@zua.edu.cn (L. Xue), yuminliu@zzu.edu.cn (Y. Liu), likailing@nuaa.edu.cn (K. Li).

https://doi.org/10.1016/j.eswa.2026.132723
Received 7 January 2026; Received in revised form 27 April 2026; Accepted 1 May 2026
Available online 3 May 2026

(Wang and Yang, Nov. (2024)) and gradient boosting decision trees (Wu et al., Apr. (2021)), were introduced. These methods demonstrate strong capability in modeling nonlinear patterns but often overlook the inherent temporal structure of the data (Lin et al., Mar. (2023)). With advances in computational power and the widespread availability of automated fare collection data, deep learning has emerged as the dominant paradigm in this field (Li et al., Feb. (2025); Ou et al., 2022). Methods such as recurrent neural networks (Nejadettehad et al., Jul. (2020)), long short-term memory networks (Zhang et al., Oct. (2024); Zhou et al., Jan. (2025)), graph neural networks (Jiang and Luo, Nov. (2022); Zhang et al., Jun. (2022)), and attention-based Transformer models (Chen et al., Apr. (2024)) have achieved notable success in capturing long-term temporal dependencies in passenger flow data (Ahmed et al., Dec. (2023)). These approaches can automatically learn high-level features and complex temporal relationships.

To overcome the limitations of single models in capturing complex features, hybrid approaches have been increasingly proposed (Wang et al., Feb. (2023); Liu et al., Feb. (2025)). For example, a combined framework integrates spectral clustering and correlation coefficient methods to classify passenger flow patterns (Huang et al., Jan. (2024)). To address the neglect of cross-period similarity in traditional approaches, the DTMGP model incorporates temporal, origin–destination spatial, frequency, and similarity characteristics (Diao et al., Mar. (2019)). To improve the performance of Prophet models during peak periods, researchers have introduced multi-component empirical decomposition, augmented with explicit morning and evening peak terms and automated hyperparameter optimization strategies (Yuan et al., Oct. (2024); Guan et al., Mar. (2026)). To mitigate cumulative errors in recursive forecasting, a ConvLSTM–GRU model has been developed that integrates multidimensional features, including passenger flow, weather, temperature, and holidays, within an encoder–decoder architecture (Zhao et al., Dec. (2022)). For feature decomposition, a Semi-Conv-Transformer model based on semi-supervised learning separates inflow data into trend and scale components (Mo et al., Jun. (2025)). Additionally, a CNN–Transformer–BiLSTM hybrid model employs CNNs for multidimensional feature extraction, applies attention mechanisms for feature weighting, and uses BiLSTM to capture bidirectional temporal dependencies. To better represent periodic dependencies, a parallel Transformer-based hybrid model has been proposed to model data across three distinct temporal scales (Bapaume et al., Aug. (2023)). To account for inherent randomness and uncertainty, interval forecasting methods have also been explored (Xie et al., Jan. (2025)). These approaches combine nonparametric kernel density estimation to characterize the probability distribution of forecasting errors and generate passenger flow intervals with specified confidence levels (Nong et al., Jul. (2025)). More recently, large language models have also been introduced into short-term passenger flow forecasting (Ma and Zhao, Oct. (2025)).

Despite these advances, current short-term metro passenger flow prediction methods still exhibit several limitations. First, single models struggle to simultaneously capture deterministic and stochastic characteristics. Traditional deterministic models focus on trends and periodic patterns but have limited capability in representing stochastic fluctuations. In contrast, data-driven models can capture complex nonlinear relationships but often lack explicit modeling of deterministic structures and physical interpretability. Second, existing hybrid models lack a systematic and coherent fusion mechanism. Feature-based approaches typically use the output of one model as the input to another, resulting in a unidirectional architecture that is highly susceptible to cumulative error propagation. Model-weighted hybrids rely on simple averaging or learned weights to combine outputs from independent models, often overlooking the intrinsic structural characteristics of passenger flow. Such approaches fail to distinguish between deterministic evolutionary patterns and stochastic noise. Deep feature fusion methods, which concatenate multidimensional feature vectors to learn complex nonlinear mappings, tend to homogenize heterogeneous information into a unified representation, leading to reduced interpretability. Overall, these approaches often lack in-depth analysis of the intrinsic data structure and fail to adopt targeted modeling strategies. As a result, such shallow integrations do not fully exploit the complementary strengths of individual models, and the resulting hybrid frameworks may still exhibit systematic biases.

To address the above issues, this paper proposes a hybrid prediction framework that integrates deterministic grey modeling with deep learning. The core concept is to decompose the time series into deterministic and stochastic components. The design of the proposed framework is theoretically grounded in Wold's Decomposition Theorem, which states that any stationary discrete-time stochastic process can be uniquely decomposed into two uncorrelated components: a purely deterministic part and a purely stochastic part. Based on this principle, different modeling strategies are employed for precise characterization, and forecasting accuracy is further enhanced through intelligent correction mechanisms. First, the grey fourier model (GFM) is used to model the deterministic component and periodic patterns of passenger flow. Second, for stochastic fluctuation components, an attention-based Transformer model is developed to capture nonlinear relationships and long-term dependencies. Finally, a Markov chain is introduced to characterize the state-transition behavior of residuals, enabling intelligent correction of the hybrid forecasts. The main contributions of this paper are as follows:

(1) A deterministic–stochastic hybrid modeling framework is proposed. By addressing the strong periodicity, nonlinear fluctuations, and stochastic disturbances in short-term passenger flow data, the time series is decomposed into deterministic and stochastic components to enable precise extraction of multi-level features.

(2) An integration of grey system theory and deep learning is achieved. The combination of GFM and the Transformer model preserves both the interpretability and theoretical foundation of deterministic modeling and the powerful representation capability of deep learning. The incorporation of Markov chain-based residual state modeling further enhances forecasting accuracy through intelligent optimization.

(3) A novel solution for urban rail transit passenger flow forecasting is provided. Experimental results based on real-world data demonstrate that the proposed method exhibits strong robustness and generalization ability across diverse passenger flow patterns, with significantly improved accuracy compared to single models.

The remainder of this paper is organized as follows: Section 2 introduces GFM-based deterministic component modeling. Section 3 presents Transformer-based stochastic fluctuation modeling, and Section 4 describes Markov chain-based residual correction. Section 5 provides a case study, including results analysis, ablation experiments, and comparative evaluation. Finally, Section 6 summarizes the study and outlines directions for future research.

2. GFM-based deterministic component modeling

Grey system theory provides an effective framework for analyzing systems characterized by partially known and partially unknown information (Deng, Mar. (1982)). In particular, GFM extends this framework by incorporating a Fourier series to represent periodic fluctuations, thereby enabling the simultaneous characterization of both trend and periodic components in time series data (Zhou and Ding, Feb. (2021); Wang et al., 2022).

2.1. Model construction

Definition 1. (*For a time series $X(t) = \{x(t_1), x(t_2), \cdots, x(t_m)\}$, the differential equation of the traditional grey model GM(1,1) is*)

$$\frac{d}{dt}y(t) = \alpha y(t) + \beta \tag{1}$$

where $Y(t) = \{y(t_1), y(t_2), \cdots, y(t_m)\}$ is the accumulated generating *sequence, and $y(t_k)$ satisfies*

$$y(t_k)=\sum_{i=1}^{k}h_ix(t_i),h_k=\begin{cases}1, & k=1\\ t_k-t_{k-1}, & k=2,3,\cdots,m\end{cases} \tag{2}$$

Eq. (1) describes the evolution pattern of y. From the model solution, it can be observed that, in the GM(1,1) model, the development of y follows an exponential law (Li and Xie, Oct. (2023)). However, under the influence of external environmental factors, many natural phenomena and socio-economic processes exhibit pronounced periodic variations (Yin, Mar. (2024)). To account for such external effects, a periodic function f is introduced to characterize external influences.

Definition 2. (*Assuming the angular frequency of the periodic time series $X(t)$ is ω, the GFM is defined as*)

$$\begin{cases}\dfrac{d}{dt}x(t)=\alpha x(t)+a_0+\sum_{n=1}^{N}(a_n\cos(n\omega t)+b_n\sin(n\omega t))\\ x(t_1)=\eta\end{cases} \tag{3}$$

abbreviated as GFM, where α and $a_0,a_1,b_1,a_2,b_2,\cdots,a_N,b_N$ are model parameters, η is the initial value, and N is the adjustable nonlinear parameter.

Here, to avoid the use of the inverse accumulated generating operator, the original data are directly employed for modeling. For periodic time series, the angular frequency can be readily determined.

Theorem 1. (*The analytical solution of the GFM is*)

$$x(t)=ce^{\alpha t}+A_0+\sum_{n=1}^{N}(A_n\cos(n\omega t)+B_n\sin(n\omega t)) \tag{4}$$

where

$$\Theta=T^{-1}\theta \tag{5}$$

$$c=e^{-\alpha t_1}\left[\eta-A_0-\sum_{n=1}^{N}(A_n\cos(n\omega t_1)+B_n\sin(n\omega t_1))\right] \tag{6}$$

$$\Theta=\begin{bmatrix}A_0\\A_1\\B_1\\\vdots\\A_N\\B_N\end{bmatrix},T=\begin{bmatrix}-\alpha&0&0&\cdots&0&0\\0&-\alpha&\omega&\cdots&0&0\\0&-\omega&-\alpha&\cdots&0&0\\\vdots&\vdots&\vdots&\ddots&\vdots&\vdots\\0&0&0&\cdots&-\alpha&N\omega\\0&0&0&\cdots&-N\omega&-\alpha\end{bmatrix},\theta=\begin{bmatrix}a_0\\a_1\\b_1\\\vdots\\a_N\\b_N\end{bmatrix} \tag{7}$$

2.2. Parameter estimation

To identify the unknown structural parameters, a parameter estimation scheme combining integral transformation and the least squares method is adopted.

Applying the integration operator to Eq. (3) over the interval $[t_1,t]$ yields:

$$\int_{t_1}^{t}dx(\tau)=\alpha\int_{t_1}^{t}x(\tau)d\tau+\int_{t_1}^{t}F(\tau)d\tau \tag{8}$$

where $F(t)=a_0+\sum_{n=1}^{N}(a_n\cos(n\omega t)+b_n\sin(n\omega t))$. By invoking the fundamental theorem of calculus, Eq. (8) can be reformulated as follows:

$$x(t)-x(t_1)=\alpha\int_{t_1}^{t}x(\tau)d\tau+[\mathscr{F}(t)-\mathscr{F}(t_1)] \tag{9}$$

Here, $\mathscr{F}(t)=a_0t+\sum_{n=1}^{N}\left[\frac{a_n}{n\omega}\sin(n\omega t)-\frac{b_n}{n\omega}\cos(n\omega t)\right]+d$, with d denotes the integration constant.

For numerical implementation, the integral term $\int_{t_1}^{t}x(\tau)d\tau$ is approximated using the trapezoidal rule:

$$z(t_k)\triangleq\int_{t_1}^{t_k}x(\tau)d\tau=\frac{1}{2}\sum_{i=2}^{k}h_ix(t_{i-1})+\frac{1}{2}\sum_{i=2}^{k}h_ix(t_i),k=2,3,\cdots,m \tag{10}$$

By incorporating the background value $z(t_k)$ into Eq. (9) and discretizing the remaining terms, the corresponding discrete equations are obtained:

$$x(t_k)=\alpha z(t_k)+\eta+[\mathscr{F}(t_k)-\mathscr{F}(t_1)]+\varepsilon(t_k),k=2,3,\cdots,m \tag{11}$$

where $\varepsilon(t_k)$ represents the cumulative errors arising from discretization and noise. Consequently, the resulting $m-1$ equations can be expressed in compact matrix form:

$$X=H\Xi+\varepsilon \tag{12}$$

where

$$\Xi=\begin{bmatrix}\alpha\\\eta\\\theta\end{bmatrix},X=\begin{bmatrix}x(t_2)\\x(t_3)\\\vdots\\x(t_m)\end{bmatrix} \tag{13}$$

$$H=\begin{bmatrix}z(t_2)&1&t_2-t_1&s_1(t_2)&c_1(t_2)&\cdots&s_N(t_2)&c_N(t_2)\\z(t_3)&1&t_3-t_1&s_1(t_3)&c_1(t_3)&\cdots&s_N(t_3)&c_N(t_3)\\\vdots&\vdots&\vdots&\vdots&\vdots&\ddots&\vdots&\vdots\\z(t_m)&1&t_m-t_1&s_1(t_m)&c_1(t_m)&\cdots&s_N(t_m)&c_N(t_m)\end{bmatrix},\varepsilon=\begin{bmatrix}\varepsilon(t_2)\\\varepsilon(t_3)\\\vdots\\\varepsilon(t_m)\end{bmatrix} \tag{14}$$

and

$$s_n(t)=\frac{\sin(n\omega t)-\sin(n\omega t_1)}{n\omega},c_n(t)=\frac{-\cos(n\omega t)+\cos(n\omega t_1)}{n\omega} \tag{15}$$

Theorem 2. (*The optimal parameter estimate $\widehat{\Xi}$ is obtained by the squared Euclidean norm of the residual vector, $L(\Xi)=\|X-H\Xi\|_2^2$:*)

$$\widehat{\Xi}=(H^TH)^{-1}H^TX \tag{16}$$

The corresponding estimated time response function can then be obtained as follows:

$$\widehat{x}(t)=\widehat{c}e^{\widehat{\alpha}t}+\widehat{A}_0+\sum_{n=1}^{N}(\widehat{A}_n\cos(n\omega t)+\widehat{B}_n\sin(n\omega t)) \tag{17}$$

Predictions are subsequently generated by extrapolating this function to future time points $\{t_k\}_{k=1}^{m+p}$, where p denotes the number of forecasting steps.

2.3. Determination of Fourier order

The order N of the Fourier series is a critical hyperparameter in the GFM. A small value of N may lead to underfitting, as it fails to capture complex waveform characteristics, whereas an excessively large N increases the risk of overfitting. Therefore, an appropriate selection of N is essential to balance model flexibility and generalization ability.

Let h, T, and $f_s=1/h$ denote the sampling interval, period, and sampling frequency, respectively. To satisfy the Nyquist–Shannon sampling theorem, the maximum resolvable frequency $f_h=Nf=N/T$ must not exceed half of the sampling frequency, i.e., $f_s\geq 2f_h$. This yields a theoretical upper bound for the model order: $N\leq T/(2h)$. Consequently, the feasible range of the order is defined as $\mathscr{N}=\{0,1,2,\cdots,\lfloor T/(2h)\rfloor\}$. The optimal order N^* is then determined by minimizing the mean absolute percentage error (MAPE) over the validation set.

3. Transformer-based fluctuation component modeling

3.1. Fluctuation sequence extraction

The GFM model can effectively capture deterministic components, including exponential trends and periodic fluctuations. However, actual passenger flow data still contain substantial irregular variations caused by random events, sudden disruptions, and nonlinear interactions, which are difficult to represent using purely deterministic models (Xue et al., Feb. (2022); Xu et al., Jan. (2024)). Therefore, a Transformer-based model is introduced to capture these fluctuation characteristics. First, the original passenger flow sequence is decomposed as follows:

$$X_t^{original} = X_t^{trend+period} + X_t^{fluctuation} \tag{18}$$

where $X_t^{original}$ denotes the original passenger flow, $X_t^{trend+period}$ represents the deterministic component fitted by the GFM model, and $X_t^{fluctuation}$ denotes the fluctuation component. Based on the fitted values obtained from the GFM model in Eq. (17), the fluctuation sequence is defined as follows:

$$X_t^{fluctuation} = X_t^{original} - \widehat{X}_t^{GFM} \tag{19}$$

The stochastic fluctuation sequence $X_t^{fluctuation}$ contains all random variation components in the original data. To eliminate scale effects and accelerate model convergence, standardization is applied to the fluctuation sequence as follows:

$$\widetilde{X}_t^{fluctuation} = \frac{X_t^{fluctuation} - \mu_{fluctuation}}{\sigma_{fluctuation}} \tag{20}$$

where $\mu_{fluctuation}$ and $\sigma_{fluctuation}$ denote the mean and standard deviation of the sequence, respectively. The Transformer model is then used to learn the temporal dependencies of the standardized sequence $\left\{\widetilde{X}_t^{fluctuation}\right\}$ and to predict fluctuations at future time steps.

3.2. Multi-feature fusion

To enhance the representation of passenger flow characteristics, a multi-feature fusion input strategy is adopted. In addition to the stochastic fluctuation sequence itself, auxiliary features are incorporated, including temporal features, statistical features (e.g., sliding window mean, standard deviation, maximum and minimum values), and differential features. The first-order difference is defined as $\Delta X_t^{fluctuation} = X_t^{fluctuation} - X_{t-1}^{fluctuation}$, which captures local variation trends. These multidimensional features are then mapped into d_{model}-dimensional embeddings via a fully connected layer:

$$\boldsymbol{E}_t = \boldsymbol{W}_e\left[\text{concat}\left(\widetilde{X}_t^{fluctuation}, \text{time}_t, \text{stat}_t, \Delta\widetilde{X}_t^{fluctuation}\right)\right] + \boldsymbol{b}_e \tag{21}$$

where $\mathbf{W}_e \in \mathbb{R}^{d_{model} \times d_{input}}$ is the learnable embedding matrix, and d_{input} denotes the dimension of the input feature vector.

Given that the Transformer model lacks a recurrent structure, it cannot inherently capture sequential order information. To address this limitation, positional encoding is introduced to enrich the input embeddings with temporal order information (Xu et al., Oct. (2023)). Given the strong periodicity of metro passenger flow, learnable periodic positional encoding is adopted instead of fixed sinusoidal encoding. This design enables the model to better capture intra-day and intra-week periodic patterns. For time step t, the positional encoding is formulated as follows:

$$\begin{aligned} PE(t,2i) &= \sin\left(\frac{t}{10000^{2i/d_{model}}}\right) + \boldsymbol{W}_{pe}^{day}[\text{hour}(t)] \\ PE(t,2i+1) &= \cos\left(\frac{t}{10000^{2i/d_{model}}}\right) + \boldsymbol{W}_{pe}^{week}[\text{weekday}(t)] \end{aligned} \tag{22}$$

where $\mathbf{W}_{pe}^{day}$ and $\mathbf{W}_{pe}^{week}$ are learnable positional embedding matrices corresponding to intra-day and intra-week periodicities, respectively. The final input representation is given by:

$$\boldsymbol{X}_{input} = \boldsymbol{E} + \boldsymbol{PE} + \text{Dropout}\left(\boldsymbol{E}, p_{drop}\right) \tag{23}$$

The use of a hybrid positional encoding is motivated by two key characteristics of metro passenger flow: stable periodicity and specific temporal anomalies. Classical sinusoidal encoding employs fixed periodic functions, providing a strong inductive bias for relative positional relationships and translation invariance. This makes it particularly effective for capturing the inherent periodic rhythms in metro passenger flow. However, because it is non-trainable, purely sinusoidal encoding lacks the flexibility to adapt to dataset-specific absolute temporal irregularities. In contrast, learnable positional embeddings are effective at capturing absolute, location-specific temporal features and can adapt to dataset characteristics. These learnable components enable the model to better capture fine-grained temporal variations within specific time contexts. By combining these two approaches, the Transformer block can simultaneously capture global periodic structures and local adaptive variations. This hybrid design therefore ensures a more comprehensive and expressive representation of temporal information.

3.3. Attention mechanism

The multi-head attention mechanism is the core component of the Transformer architecture. It enables the model to attend to information from all time steps in the historical sequence and to dynamically evaluate their relative importance for the current forecasting task.

This mechanism is implemented via scaled dot-product attention, in which each input element is linearly transformed into three vectors: Query (Q), Key (K), and Value (V), through learnable projection matrices.

$$\boldsymbol{Q} = \boldsymbol{X}_{input}\boldsymbol{W}_Q, \boldsymbol{K} = \boldsymbol{X}_{input}\boldsymbol{W}_K, \boldsymbol{V} = \boldsymbol{X}_{input}\boldsymbol{W}_V \tag{24}$$

Attention scores are obtained by computing the dot product between the query vector and all key vectors. To prevent excessively large dot-product values that may lead to vanishing gradients, a scaling factor is applied. The resulting scores are then normalized into attention weights using the Softmax function, and these weights are used to compute a weighted sum of the corresponding value vectors:

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\mathbf{QK}^{\mathrm{T}}}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{25}$$

After each attention sublayer, a position-wise feed-forward network is applied. The main role of this feed-forward network is to introduce nonlinear transformations, further processing the representations generated by the attention mechanism and enhancing the model's ability to capture complex fluctuation patterns.

After passing through multiple encoder layers, the model obtains deep feature representations for each time step in the input sequence $\mathbf{H} = \{\mathbf{h}_1, \mathbf{h}_2, \cdots, \mathbf{h}_L\}$, where L denotes the input sequence length. Given that the objective is to forecast passenger flow fluctuations over the next τ time steps, an autoregressive decoding strategy is adopted, where the prediction from the previous step is used as input for the next step:

$$\widehat{y}_{t+k}^{(fluctuation)} = f_{decoder}\left(\boldsymbol{H}, \widehat{y}_{t+1}^{(fluctuation)}, \cdots, \widehat{y}_{t+k-1}^{(fluctuation)}\right) \tag{26}$$

where $f_{decoder}(\cdot)$ denotes the decoder function, and $k = 1, 2, \cdots, \tau$ represents the forecasting horizon.

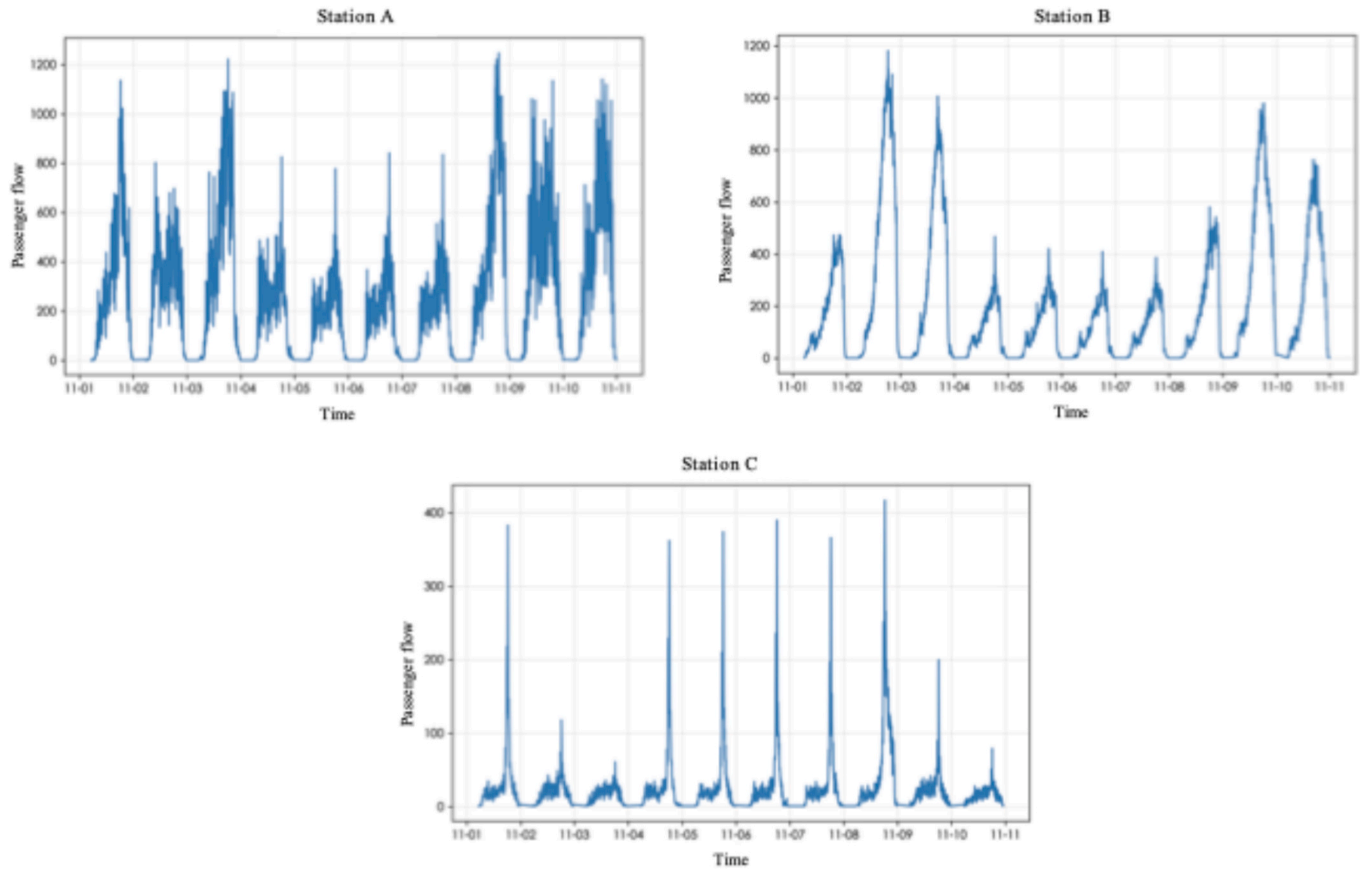

Fig. 1. Visualization of passenger flow at metro stations

4. Markov chain-based intelligent residual correction

Based on the aforementioned modeling results, two complementary forecasting components are obtained: the deterministic component $\hat{y}_t^{(GFM)}$ generated by the GFM model and the stochastic fluctuation component $\hat{y}_t^{(Trans)}$ produced by the Transformer model. Although the GFM and Transformer separately capture the deterministic patterns and primary stochastic variations in passenger flow data, residual systematic biases and weak signals may persist in the final forecasts. These residuals may arise from rare events such as extreme weather and sudden disruptions, approximation errors inherent in the model structures, and cumulative effects of measurement noise (Ye et al., Sep. (2022)). To further improve forecasting accuracy, a Markov chain is introduced to perform intelligent correction of the residuals.

Based on the distribution characteristics of historical residuals $e_t = y_t - [\hat{y}_t^{(GFM)} + \hat{y}_t^{(Trans)}]$, the value range is discretized into M non-overlapping state intervals. In this study, an equal-interval partitioning strategy is adopted to ensure clear state boundaries and computational simplicity.

First, the range of residuals in the training set is determined as follows: $e_{\min} = \min_{t=1}^{T_{train}} e_t, e_{\max} = \max_{t=1}^{T_{train}} e_t$. Next, the state interval width is calculated as follows: $\Delta e = \frac{e_{\max} - e_{\min}}{M}$ Accordingly, the j-th state interval is defined as follows: $S_j = [e_{\min} + (j-1)\Delta e, e_{\min} + j\Delta e), j = 1, 2, \cdots, M-1$. For the final interval, a closed form is adopted to ensure inclusion of boundary values: $S_M = [e_{\min} + (M-1)\Delta e, e_{\max}]$.

Define the state mapping function:

$$s_t = \mathscr{S}(e_t) = \begin{cases} \left\lceil \frac{e_t - e_{\min}}{\Delta e} \right\rceil, & \text{if } e_t < e_{\max} \\ M, & \text{if } e_t = e_{\max} \end{cases} \tag{27}$$

where $\lceil \cdot \rceil$ denotes the ceiling operator. For convenience in subsequent correction calculations, the representative value of each state is defined as the midpoint of the corresponding interval:

$$\bar{e}_j = e_{\min} + (j - 0.5)\Delta e, j = 1, 2, \cdots, M \tag{28}$$

The number of states M is a key hyperparameter that significantly influences the effectiveness of Markov-based correction. Based on the residual state sequence $\{s_t\}_{t=1}^{T_{train}}$, the frequencies n_{ij}, representing transitions from state S_i to state S_j, are counted to construct the transition probability matrix:

$$\boldsymbol{P} = \left[p_{ij}\right]_{M \times M}, p_{ij} = P\left(s_t = S_j \middle| s_{t-1} = S_i\right) = \frac{n_{ij}}{\sum_{k=1}^{M} n_{ik}} \tag{29}$$

where the transition frequency n_{ij} is defined as follows:

$$n_{ij} = \sum_{t=2}^{T_{train}} \mathbb{I}(s_{t-1} = i \text{ and } s_t = j) \tag{30}$$

Here, $\mathbb{I}(\cdot)$ denotes the indicator function, which equals 1 when the condition is satisfied and 0 otherwise. The transition probability p_{ij} represents the probability that, given the residual at time $t-1$ is in state S_i, the residual at time t transitions to state S_j. This transition matrix satisfies the fundamental property of Markov chains: $\sum_{j=1}^{M} p_{ij} = 1, \forall i \in \{1, 2, \cdots, M\}$. If a certain state S_i does not appear in the training set (i.e., $\sum_{k=1}^{M} n_{ik} = 0$), the transition probabilities for that state cannot be estimated using frequency statistics. In this case, Laplace smoothing is applied:

$$p_{ij} = \frac{n_{ij} + \alpha}{\sum_{k=1}^{M} (n_{ik} + \alpha)} = \frac{n_{ij} + \alpha}{\sum_{k=1}^{M} n_{ik} + M\alpha} \tag{31}$$

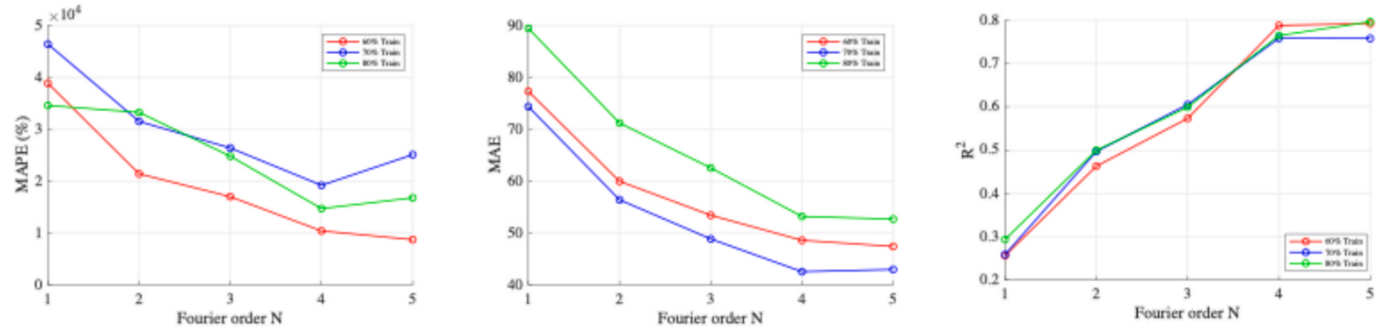

Fig. 2. The effects of different N under varying data lengths

Here, the smoothing parameter α is typically set to a small value (e.g., $\alpha = 0.1$) to avoid zero-probability issues while minimally affecting the empirical distribution of observed data. When $\sum_{k=1}^{M} n_{ik} = 0$, the above formula naturally reduces to a uniform distribution, i.e., $p_{ij} = 1/M$.

In the forecasting phase, assuming the residual e_{t-1} at time $t-1$ is known and its corresponding state is $s_{t-1} = S_i$, the conditional expectation of the residual at time t can be estimated as follows:

$$\tilde{e}_t^{(Markov)} = \mathbb{E}[e_t | s_{t-1} = S_i] = \sum_{j=1}^{M} p_{ij} \cdot \bar{e}_j \tag{32}$$

where $\bar{e}_j$ denotes the representative value of state S_j.

5. Case study

5.1. Data collection

This study uses real passenger flow data collected from November 1 to November 8, 2024. The dataset covers a complete weekly cycle, enabling the evaluation of the model under pronounced transitions between weekday and weekend travel patterns, which is a critical scenario in urban transit operations. Three metro stations with representative operational characteristics are selected for model training and validation. The data sampling interval is 15 minutes, resulting in 96 time steps per day.

Figure 1 illustrates the temporal variation characteristics of passenger flow across the three selected stations. Overall, all stations exhibit pronounced daily and weekly periodicity. Passenger flow during nighttime (0:00–6:00) is close to zero, whereas daytime periods show varying degrees of fluctuations and peak values. In addition to the deterministic periodic components, all stations also present substantial short-term stochastic fluctuations as well as sporadic extreme passenger flow events. However, the passenger flow characteristics differ significantly among the three stations, particularly in terms of flow magnitude, peak-to-valley ratios, fluctuation regularity, and weekend effects. These differences reflect typical passenger flow patterns across stations with different geographical and functional attributes. Overall, these characteristics provide rich validation scenarios for the proposed modeling framework and establish a diversified experimental basis for comprehensively evaluating its effectiveness, robustness, and generalization capability.

5.2. Evaluation metrics

The forecasting results are evaluated using mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2) as performance metrics. The corresponding formulas are as follows:

$$MAE = \frac{1}{n} \sum_{t=1}^{n} |x_t - \hat{x}_t| \tag{33}$$

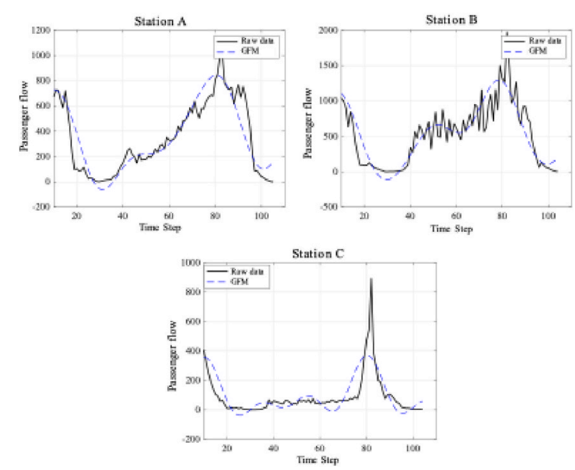

Fig. 3. GFM modeling performance

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^{n} (x_t - \hat{x}_t)^2} \tag{34}$$

$$R^2 = 1 - \frac{\sum_{t=1}^{n} (x_t - \hat{x}_t)^2}{\sum_{t=1}^{n} (x_t - \bar{x})^2} \tag{35}$$

In the above expressions, n denotes the total number of passenger flow time steps, x_t represents the observed (true) passenger flow at time step t, $\hat{x}_t$ denotes the predicted value of passenger flow at time step t, and $\bar{x}$ is the mean of observed passenger flow values.

5.3. Results analysis

To further demonstrate the robustness of the Fourier order selection, an effect analysis is conducted by varying the training set split ratio from 60% to 80%. As illustrated in Figure 2, although the MAPE exhibits slight fluctuations under different training sizes, the optimal value of N consistently remains at 4 across all scenarios. Meanwhile, R^2 increases steadily as N grows from 1 to 4, but no further improvement is observed once $N > 4$. This indicates that $N = 4$ serves as a critical threshold that sufficiently captures the dominant periodic patterns of passenger flow, while additional model complexity does not yield further performance gains. The observed stability suggests that N is an intrinsic characteristic parameter of the periodic structure of passenger flow rather than a hyperparameter sensitive to specific data partitions. This finding further confirms the reliability and robustness of the proposed GFM component.

To validate the capability of the GFM model in capturing deterministic components, Figure 3 presents the modeling results for three representative metro stations. Overall, the GFM model effectively captures the main trend evolution and periodic fluctuation patterns of

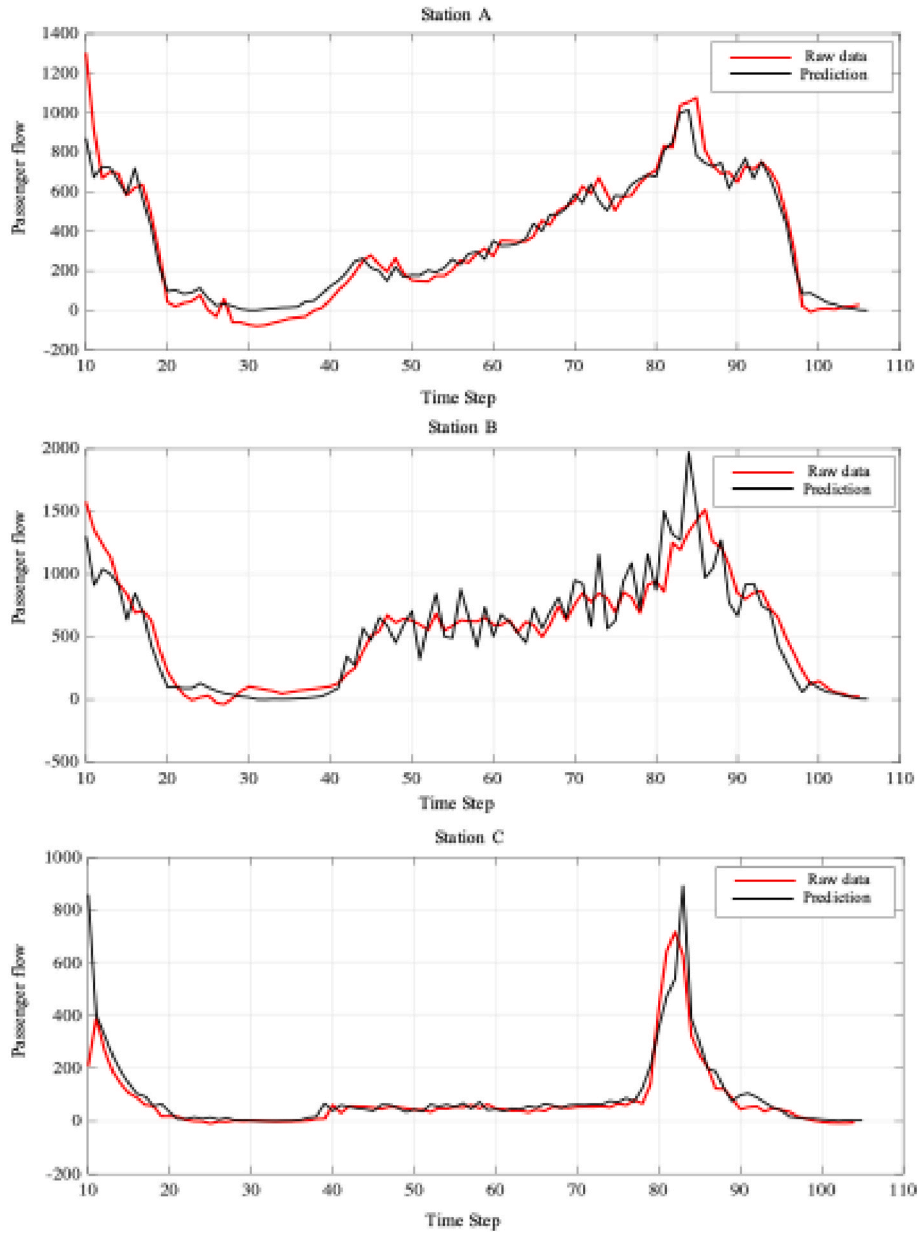

Fig. 4. Forecasting results of the proposed model

passenger flow across all three stations.

At station A, the model accurately captures the bimodal characteristics of passenger flow, with the predicted curve showing high consistency with the observed values during peak periods. Station B, as a major transportation hub, exhibits a more complex pattern with continuous fluctuations. The GFM model effectively fits this periodic behavior through the superposition of multi-order Fourier harmonics. In the fluctuation intervals of time steps 20–40 and 80–100, the predicted values closely follow the observed variations. At station C, passenger flow remains at a relatively low level during time steps 1–70, followed by a sharp peak around time step 80. The GFM model successfully captures this extreme distribution pattern through the combined representation of exponential trend and periodic components. Its predictions closely match the observed values during low-flow periods and accurately reproduce both the timing and magnitude of the peak. However, certain local discrepancies can still be observed. For instance, in the interval of time steps 40–60 at station A, the GFM prediction curves are relatively smooth and fail to fully capture short-term stochastic disturbances. Similarly, at station B during the moderate flow interval of time steps 30–50, the model slightly underestimates the actual passenger flow. These results indicate that during off-peak periods in complex transportation hubs, relying solely on deterministic modeling is insufficient to fully characterize passenger flow dynamics influenced by multiple stochastic factors.

These observations further validate the rationality of the modeling framework proposed in this study. As a deterministic modeling approach, the GFM model can effectively extract trend information and dominant periodic components from passenger flow data; however, its ability to capture short-term stochastic fluctuations and nonlinear disturbances remains limited. This limitation further highlights the necessity of introducing the Transformer model to characterize random fluctuation components.

Figure 4 presents the forecasting results of the proposed hybrid model at three representative stations, while Figure 5 reports the corresponding accuracy evaluation. Overall, the proposed model achieves high prediction accuracy across all stations. Compared with the GFM results shown in Figure 3, the predicted curves of the hybrid model exhibit a much closer alignment with the true values, indicating a

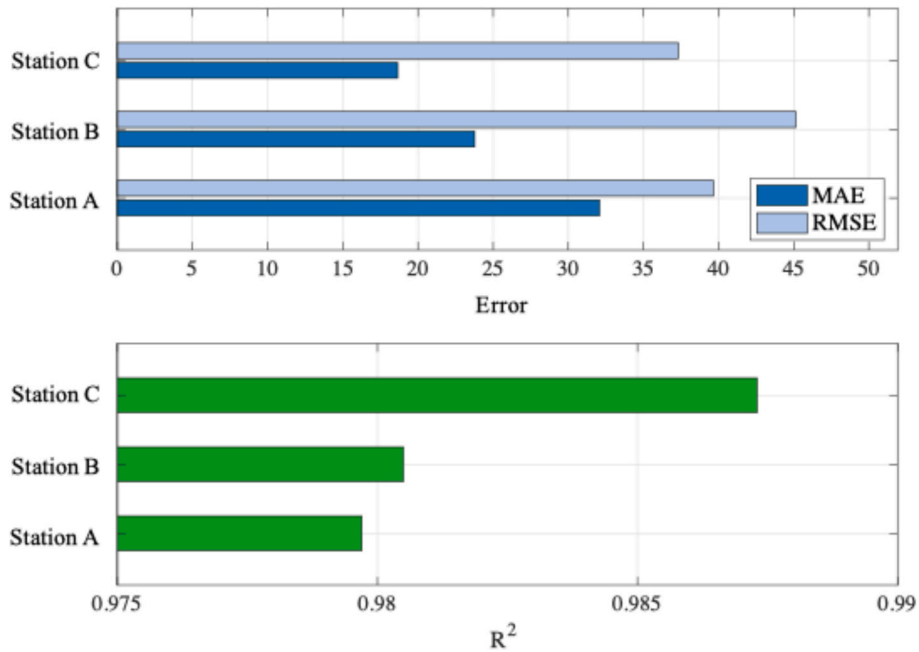

Fig. 5. Accuracy evaluation of the proposed model

Table 1
Performance of component models

	MAE	RMSE	R^2
GFM	86.96	122.77	0.8299
Transformer	61.13	72.91	0.9314
GFM+Markov	51.60	70.66	0.9440
Transformer + Markov	34.07	46.27	0.9724
Transformer + GFM	55.22	70.28	0.9363
Proposed	32.08	39.65	0.9797

significant improvement in forecasting performance. In addition, the predicted trajectories at all three stations remain smooth, without abrupt changes or excessive oscillations, reflecting the model's stability and robustness. From a more detailed perspective, after incorporating the Transformer module, the model demonstrates strong capability in capturing short-term fluctuations. Taking station A as an example, the passenger flow within the time interval 50–70 exhibits high-frequency oscillations, which may be caused by uncertain factors such as random arrival bursts and temporary passenger aggregation. The Transformer effectively captures these nonlinear variations by learning temporal dependencies through the attention mechanism, resulting in predicted oscillations that are highly consistent with the observed amplitude and frequency. Similarly, in the time step interval 30–50 at station B, the GFM model shows a tendency to underestimate passenger flow, whereas the hybrid model effectively mitigates this systematic bias through the combined effect of Transformer-based fluctuation modeling and Markov chain-based residual correction, achieving a closer agreement between predicted and actual values. These improvements clearly demonstrate the effectiveness and superiority of the proposed modeling framework.

In summary, the hybrid forecasting framework proposed in this study demonstrates superior performance across different station types and passenger flow patterns. The model not only accurately captures long-term trends and periodic characteristics of passenger flow but also effectively characterizes short-term stochastic fluctuations and systematic biases, thereby achieving comprehensive and high-precision short-term metro passenger flow forecasting.

5.4. Ablation study

To comprehensively evaluate the contribution of each component in the proposed hybrid framework, this section conducts ablation experiments. By progressively incorporating different modeling modules, the incremental impact of each component on overall performance is analyzed. The results are reported in Table 1 and Figure 6.

The GFM model, serving as the baseline deterministic approach, achieves MAE = 86.96, $RMSE$ = 122.77, and R^2=0.8299 across the three evaluation metrics. Although GFM effectively captured trends and

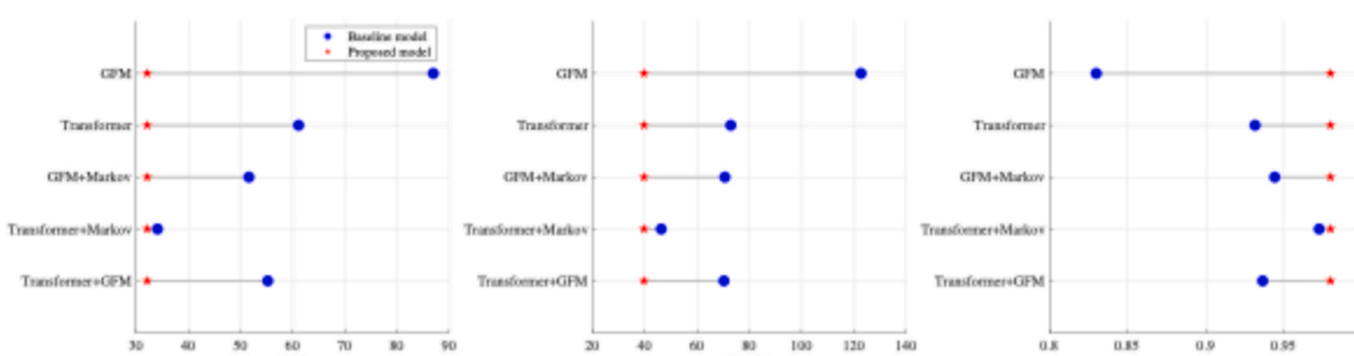

Fig. 6. Accuracy comparison of component models

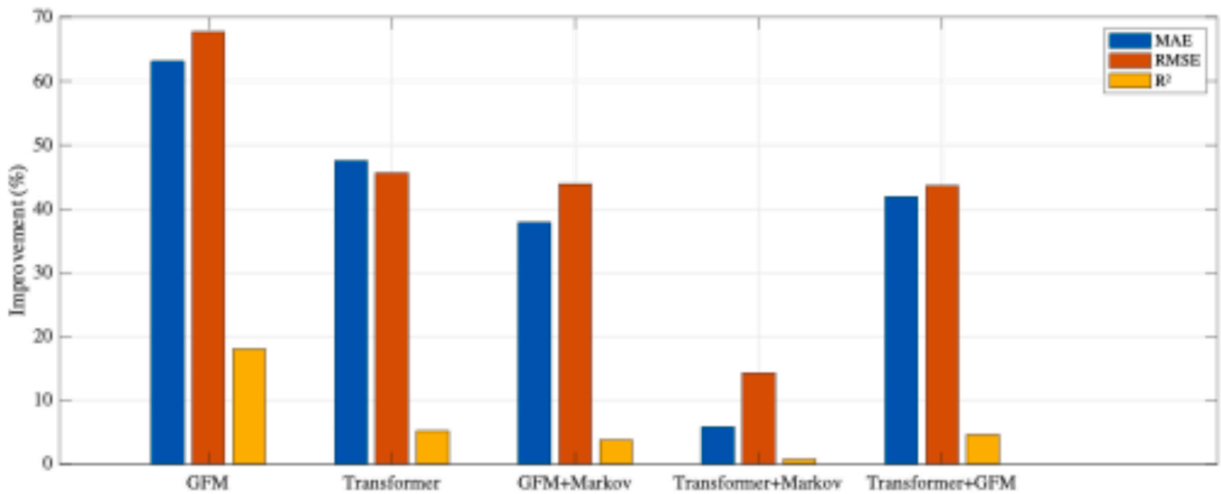

Fig. 7. Accuracy improvement

periodic features of passenger flow, its prediction error remains relatively high due to its inability to model stochastic fluctuations. The R^2 value of 0.8299 indicates that 83% of the variance in the data is explained by the model, suggesting that deterministic components play a dominant role in passenger flow dynamics and providing a solid foundation for subsequent modeling. When the Transformer is applied independently to the raw passenger flow sequence, it achieves significantly better performance than the GFM model. Compared with GFM, the Transformer reduces MAE by 29.7% and RMSE by 40.6%, while improving R^2 by 10.15%. These results demonstrate the strong capability of deep learning methods in capturing complex nonlinear temporal dependencies. However, despite its overall superiority, the standalone Transformer still leaves room for further improvement.

The combination of GFM and Markov chain achieves a significant improvement over the standalone GFM model, indicating that Markov-based modeling of residual state transitions can effectively correct systematic biases in deterministic forecasting. The combination of Transformer and Markov chain achieves the best performance among the two-component hybrid models. The GFM–Transformer combination also shows clear improvement over the GFM baseline; however, it remains slightly inferior to the Transformer–Markov framework. This result suggests that simply combining two modeling approaches does not necessarily yield optimal performance; instead, the effectiveness of the hybrid model depends on a well-designed integration strategy. The relatively weaker performance of the GFM–Transformer combination may be attributed to the limited size and low signal-to-noise ratio of the fluctuation sequence, which constrains the learning capability of the Transformer. To further investigate this phenomenon, the statistical properties of the data before and after GFM decomposition are analyzed. It is observed that after removing the deterministic trend component via GFM, the residual fluctuation sequence becomes considerably more irregular. Specifically, the coefficient of variation increases sharply from 1.49 to 204.34, while the first-order autocorrelation decreases from 0.84 to 0.57. These changes indicate that the residual series is dominated by noise and exhibits weaker temporal dependence. Given that the Transformer's attention mechanism relies on identifying stable and repetitive patterns for effective learning, its performance degrades when applied to such highly irregular residuals. This explains why incorporating a Markov chain is necessary, as it is specifically designed to model and correct stochastic state transitions that are difficult for the Transformer to capture alone.

From the scatter plot comparison in Figure 6, it can be observed that the proposed model significantly outperforms all baseline and component models across the three evaluation metrics, achieving the best overall forecasting performance. The proposed framework consistently demonstrates superior accuracy under all ablation settings. Compared with the strongest competing model (Transformer + Markov), the proposed model still achieves a 5.8% reduction in MAE, a 14.3% reduction in RMSE, and a 0.73% improvement in R^2. From the accuracy improvement results shown in Figure 7, the performance gains of different combination strategies relative to each component model can be more intuitively observed. Overall, the experimental results fully demonstrate that, compared with single methods or simple hybrid strategies, the modeling framework proposed in this study can more comprehensively capture multi-level features of passenger flow data, thereby significantly improving forecasting accuracy and robustness.

Table 2
Performance evaluation results of comparison models

Model	MAE	RMSE	R^2
Random Forest	38.41	58.24	0.9617
Gradient Boosting	41.55	59.80	0.9596
MLP Regressor	66.22	98.98	0.8894
SVR	79.58	130.95	0.8064
ARIMA	265.63	297.10	-
Holt–Winters	257.10	346.23	-
SARIMA	277.64	348.09	-
Proposed	32.08	39.65	0.9797

5.5. Comparative analysis

To comprehensively evaluate the superiority of the proposed method, comparative experiments are conducted against several mainstream forecasting approaches. Both traditional time series models (ARIMA, Holt–Winters, SARIMA) and machine learning models (Random Forest, Gradient Boosting, SVR, MLP Regressor) are selected for comparison.

As shown in Table 2, traditional time series methods (ARIMA, Holt–Winters, SARIMA) perform poorly in the short-term metro passenger flow forecasting task. Their errors are significantly higher than those of the proposed method, and reliable R^2 values cannot be obtained. This indicates that traditional time series models struggle to capture complex nonlinear relationships and multi-period nested structures in passenger flow data. Moreover, these methods impose strict assumptions on data stationarity, whereas real-world passenger flow data exhibit strong non-stationary behavior.

Figure 8 illustrates a clear performance gap between traditional methods and the proposed approach across the three evaluation metrics. Machine learning methods show substantially better performance than traditional time series models. Although Random Forest and Gradient

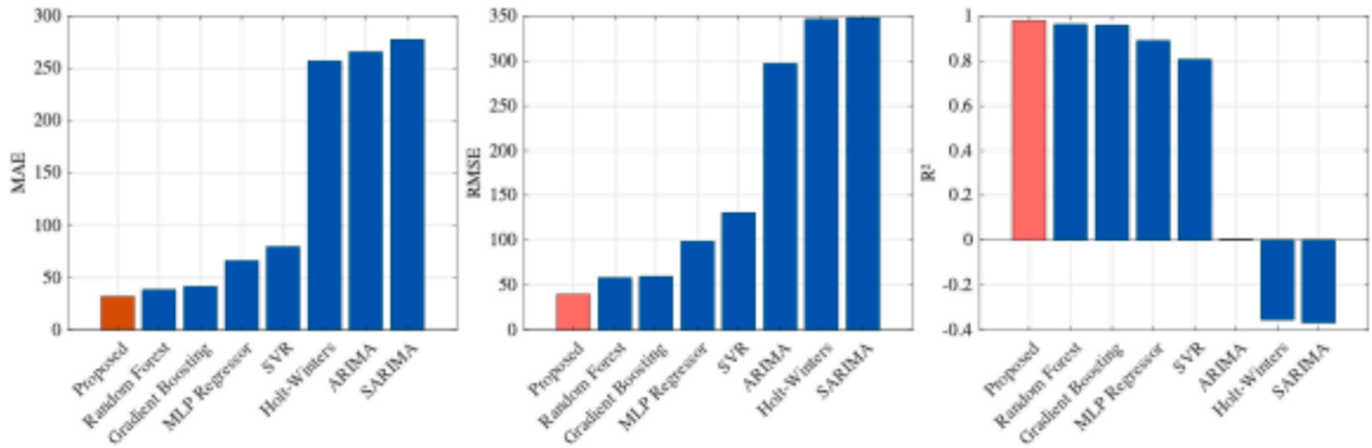

Fig. 8. Accuracy comparison with other models

Boosting achieve relatively strong forecasting results, they still exhibit noticeable performance gaps compared with the proposed method. From the bar chart comparison in Figure 8, it is evident that the proposed method (red bars) yields significantly lower MAE and RMSE values than all machine learning baselines (blue bars), while achieving a substantially higher R^2. These results confirm that the proposed hybrid framework effectively improves prediction accuracy.

Beyond standard error-based metrics, practical deployment in intelligent urban rail transit systems requires a more comprehensive evaluation of operational value. To this end, two application-oriented metrics are introduced: computational efficiency and peak-hour prediction accuracy. (1) Computational efficiency. Although inference time per step for traditional machine learning models such as SVR (0.0126 s) and Random Forest (0.0012 s) is relatively higher, the proposed method maintains high efficiency with an average inference time of 0.0004 s. In a real-world deployment scenario, the total end-to-end processing time, including data preprocessing and model inference, is only a few seconds. Compared with the 15-minute sampling interval, this delay is negligible. Therefore, the proposed framework provides sufficient decision-making time for metro operators, meeting the requirements for real-time scheduling and proactive safety management. (2) Operational safety. In metro systems, operational safety is closely related to crowd management during surge periods. Extreme passenger flows may lead to station congestion and safety risks. Therefore, performance during peak periods is a critical indicator of practical value. To evaluate this, passenger surge periods were isolated and prediction accuracy was computed separately. Despite the high volatility and extreme values in these periods, the proposed model maintains a stable MAPE of 15%. This demonstrates that the model can reliably capture surge dynamics and provide early warning capability, offering valuable lead time for implementing crowd control and safety interventions.

6. Conclusion

While traditional expert systems rely on explicit rule-based logic, modern urban transit management requires data-driven intelligent systems capable of learning from complex dynamics. The proposed framework can be regarded as a hybrid intelligent modeling approach that addresses the key characteristics of short-term metro passenger flow, including strong periodicity, nonlinear fluctuations, and stochastic disturbances. It enables the effective extraction and representation of multi-level features in passenger flow data. The synergistic integration of the GFM and Transformer bridges two distinct modeling paradigms: time series modeling and black-box theory. In addition, the Markov chain-based modeling of residual state transitions further enhances the framework by enabling intelligent correction and improving overall forecasting accuracy. Based on real-world passenger flow data, the proposed framework demonstrates stable and reliable performance across different stations. Ablation studies further confirm the necessity and effectiveness of each component.

Despite its strong performance, several limitations remain. The deterministic trend captured by GFM may still be affected by extreme external shocks. In addition, the Markov correction mechanism lacks adaptability to dynamic residual behavior across different time periods. Therefore, future research will focus on developing more adaptive intelligent forecasting systems capable of handling varying temporal granularities and heterogeneous operational scenarios. Furthermore, extended datasets covering longer time spans, including seasonal variations and major holidays, will be incorporated to further evaluate the robustness of the model under diverse environmental conditions.

CRediT authorship contribution statement

Di Zhao: Conceptualization, Methodology, Writing – original draft. **Li Xue:** Data curation, Validation, Funding acquisition. **Yumin Liu:** Visualization, Writing – review & editing, Funding acquisition. **Kailing Li:** Supervision, Writing – review & editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by National Natural Science Foundation of China under Grant 72471210; Key Research and Development Special Project of Henan Province, China under Grant 241111222700; Key Science and Technology Tackling Project of Henan Province, China under Grant 252102211020.

Data availability

Data will be made available on request.

References

Ahmed, S., Nielsen, I. E., Tripathi, A., Siddiqui, S., Ramachandran, R. P., & Rasool, G. (Dec. 2023). Transformers in Time-Series Analysis: A Tutorial. *Circuits, Systems, and Signal Processing, 42*(12), 7433–7466.

Bapaume, T., Come, E., Ameli, M., Roos, J., Oukhellou, L., Aug. 2023. Forecasting passenger flows and headway at train level for a public transport line: Focus on atypical situations. Transportation Research Part C: Emerging Technologies 153, 104195.

Chen, B., Yang, W., Yan, B., & Zhang, K. (Apr. 2024). An advanced airport terminal cooling load forecasting model integrating SSA and CNN-transformer. *Energy and Buildings, 309*, Article 114000.
Deng, J. (Mar 1982). Control problems of grey systems. *Systems & Control Letters, 1*(5), 288–294.
Diao, Z., Zhang, D., Wang, X., Xie, K., He, S., Lu, X., & Li, Y. (Mar. 2019). A hybrid model for short-term traffic volume prediction in massive transportation systems. *IEEE Transactions on Intelligent Transportation Systems, 20*(3), 935–946.
Geng, J., Zhang, C., Yang, L., Meng, F., & Qi, J. (Mar. 2024). Integrated scheduling of metro trains and shuttle buses with passenger flow control strategy on an oversaturated metro line. *Computers & Industrial Engineering, 189*, Article 109980.
Guan, X., Li, M. Z., Qin, J., & Wang, C. (Mar. 2026). Short-term high-speed rail passenger flow forecasting integrated extended empirical mode decomposition with multivariate and bidirectional support vector machine. *Expert Systems with Applications, 298*, Article 129870.
Hopfe, D. H., Lee, K., Yu, C., & Mar. (2024). Short-term forecasting airport passenger flow during periods of volatility: Comparative investigation of time series vs. neural network models. *Journal of Air Transport Management, 115*, Article 102525.
Huang, Z., Zheng, H., & Yang, K. (Jan. 2024). Multitype Origin-Destination (OD) Passenger Flow Prediction for Urban Rail Transit: A Deep Learning Clustering First Predicting Second Integrated Framework. *Journal of Advanced Transportation, 2024*(1), Article 6629500.
Jiang, W., & Luo, J. (Nov. 2022). Graph neural network for traffic forecasting: A survey. *Expert Systems with Applications, 207*, Article 117921.
Li, H., Fu, W., Zhang, H., Liu, W., Sun, S., & Zhang, T. (Feb. 2025). Spatio–temporal graph hierarchical learning framework for metro passenger flow prediction across stations and lines. *Knowledge-Based Systems, 311*, Article 113132.
Li, J., Huang, Y., Zhang, C., & Yao, D. (Jan. 2025). How does public transport development con- tribute to carbon emission reduction? *Transportation Research Part A: Policy and Practice, 191*, Article 104327.
Li, K., & Xie, N. (Oct. 2023). Mechanism of single variable grey forecasting modelling: Integration of increment and growth rate. *Communications in Nonlinear Science and Numerical Simulation, 125*, Article 107409.
Lin, D., Broere, W., & Cui, J. (Jul. 2022). Metro systems and urban development: Impacts and implications. *Tunnelling and Underground Space Technology, 125*, Article 104509.
Lin, D., Nelson, J. D., Beecroft, M., & Cui, J. (May 2021). An overview of recent developments in China's metro systems. *Tunnelling and Underground Space Technology, 111*, Article 103783.
Lin, L., Gao, Y., Cao, B., Wang, Z., & Jia, C. (Mar. 2023). Passenger flow scale prediction of urban rail transit stations based on multilayer perceptron (MLP). *Complexity, 2023*, 1–19.
Liu, H., Wang, Z., & Fang, Z. (Feb. 2025). Integrating hybrid deep learning and path allocation for real-time inbound passenger flow prediction and anomaly detection in urban rail transit. *Information Sciences, 692*, Article 121621.
Liu, R., Li, S., & Yang, L. (Jan. 2020). Collaborative optimization for metro train scheduling and train connections combined with passenger flow control strategy. *Omega, 90*, Article 101990.
Ma, C., & Zhao, M. (Oct. 2025). Urban rail transit passenger flow prediction using large language model under multi-source spatiotemporal data fusion. *Physica A: Statistical Mechanics and its Applications, 675*, Article 130823.
Mo, Y., Zhang, J., Wang, C., Hao, X., Yang, L., & Gao, Z. (Jun. 2025). A semi-conv-transformer model for inflow prediction of newly expanding subway lines. *IEEE Transactions on Intelligent Transportation Systems, 26*(6), 8497–8512.
Nejadettehad, A., Mahini, H., & Bahrak, B. (Jul. 2020). Short-term demand forecasting for online car-hailing services using recurrent neural networks. *Applied Artificial Intelligence, 34*(9), 674–689.
Nong, X., Geng, M., & Jia, X. (Jul. 2025). Prediction of land use around urban metro stations using the CA-markov model. *Journal of Asian Architecture and Building Engineering, 24*(4), 3177–3189.
Ou, J., Sun, J., Zhu, Y., Jin, H., Liu, Y., Zhang, F., Huang, J., & Wang, X. (2022). STP-TrellisNets+: Spatial-temporal parallel TrellisNets for multi-step metro station passenger flow prediction. *IEEE Transactions on Knowledge and Data Engineering*, 1–14.
Trull, O., Garcıa-Dıaz, J. C., & Troncoso, A. (Feb. 2020). Initialization methods for multiple seasonal holt–winters forecasting models. *Mathematics, 8*(2), 268.
Wang, B., Luo, X., Wang, Z., & Su, Q. (Feb. 2023). Network-scale passenger flow forecasting methods in URT based on similarity measurement. *Journal of Transportation Engineering, Part A: Systems, 149*(2), Article 04022141.
Wang, S., & Yang, X. (Nov. 2024). Twice clustering based hybrid model for short-term passenger flow forecasting. *Transport, 39*(3), 209–228.
Wang, X., Xie, N., Yang, L., & Sep. (2022). A flexible grey Fourier model based on integral matching for forecasting seasonal PM2.5 time series. *Chaos, Solitons & Fractals, 162*, Article 112417.
Wu, W., Xia, Y., & Jin, W. (Apr. 2021). Predicting bus passenger flow and prioritizing influential factors using multi-source data: Scaled stacking gradient boosting decision trees. *IEEE Trans- actions on Intelligent Transportation Systems, 22*(4), 2510–2523.
Xie, R., Zhang, H., Li, H., Liu, W., Sun, S., & Zhang, T. (Jan. 2025). High-speed rail passenger flow prediction based on crossformer and quantile regression: A deep learning approach assisted by internet search indices. *Measurement, 242*, Article 116189.
Xu, D., Li, Y., Du, T., Zhong, H., Huang, Y., Li, L., & Xiangling, D. (Jan. 2024). Investigating the influence of outdoor temperature variations on fire-induced smoke behavior in an atrium-type underground metro station using hybrid ventilation systems. *Energy, 287*, Article 129570.
Xu, P., Zhu, X., & Clifton, D. A. (Oct. 2023). Multimodal Learning With Transformers: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 45*(10), 12113–12132.
Xue, G., Liu, S., Ren, L., Ma, Y., & Gong, D. (Feb. 2022). Forecasting the subway passenger flow under event occurrences with multivariate disturbances. *Expert Systems with Applications, 188*, Article 116057.
Xue, Q., Zhang, W., Ding, M., Yang, X., Wu, J., & Gao, Z. (Nov. 2023). Passenger flow forecasting approaches for urban rail transit: A survey. *International Journal of General Systems, 52*(8), 919–947.
Ye, J., Xu, Z., & Gou, X. (Sep. 2022). An adaptive Grey-Markov model based on parameters Self- optimization with application to passenger flow volume prediction. *Expert Systems with Applications, 202*, Article 117302.
Yin, F. (Mar. 2024). Periodic analysis of scenic spot passenger flow based on combination neural network prediction model. *Journal of Intelligent Systems, 33*(1), Article 20230158.
Yuan, Y., Jiang, X., Zhang, P., & Lai, C. S. (Oct. 2024). Short-term high-speed rail passenger flow prediction by integrating ensemble empirical mode decomposition with multivariate grey support vector machine. *Engineering Applications of Artificial Intelligence, 136*, Article 109005.
Zhang, W., Zhu, F., Lv, Y., Tan, C., Liu, W., Zhang, X., & Wang, F.-Y. (Jun. 2022). AdapGL: An adaptive graph learning algorithm for traffic prediction based on spatiotemporal neural networks. *Transportation Research Part C: Emerging Technologies, 139*, Article 103659.
Zhang, Y., Wang, X., Xie, J., & Bai, Y. (Oct. 2024). Comparative analysis of deep-learning-based models for hourly bus passenger flow forecasting. *Transportation, 51*(5), 1759–1784.
Zhao, C., Li, X., Shao, Z., Yang, H., & Wang, F. (Dec. 2022). Multi-featured spatial-temporal and dynamic multi-graph convolutional network for metro passenger flow prediction. *Connection Science, 34*(1), 1252–1272.
Zhou, W., & Ding, S. (Feb. 2021). A novel discrete grey seasonal model and its applications. *Communications in Nonlinear Science and Numerical Simulation, 93*, Article 105493.
Zhou, X., Wang, Q., Zhang, Y., Li, B., & Zhao, X. (Jan. 2025). Short-term bus passenger flow prediction based on BiLSTM neural network. *Journal of Transportation Engineering, Part A: Systems, 151*(1), Article 04024090.